

$$(367)_8 \rightarrow (?)_{10}$$

$$(19)_{10} \quad (2025)_{10} \quad 0-9$$

Number System

Humans use the **decimal** or **base-10** number system. In contrast, since computers only understand the language of 0's and 1's, they use the **binary** or **base-2** number system.

➤ Converting a number from **base-2** to **base-10**

most significant bit (msb) least significant bit (lsb)

$$(101010)_2 \rightarrow (42)_{10}$$

$$2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$$

$$32 + 0 + 8 + 0 + 2 + 0$$

Power of 2's

2^0	2^1	2^2	2^3	2^4	2^5	2^6	2^7	2^8	2^9	2^{10}	...
1	2	4	8	16	32	64	128	256	512	1024	...

$$(101010)_2 \rightarrow (42)_{10}$$

$$32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1$$

$$(101010)_2 \rightarrow (42)_{10}$$

$$32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1$$

$$(101010)_2 \rightarrow (42)_{10}$$

$$0 \times 2 + 1$$

$$= 1 \times 2 + 0$$

$$= 2 \times 2 + 1$$

$$= 5 \times 2 + 0$$

$$= 10 \times 2 + 1$$

$$= 21 \times 2 + 0$$

$$= 42$$

$$(110011)_2 \rightarrow (51)_{10}$$

$$0 \times 2 + 1$$

$$= 1 \times 2 + 1$$

$$= 3 \times 2 + 0$$

$$= 6 \times 2 + 0$$

$$= 12 \times 2 + 1$$

$$= 25 \times 2 + 1$$

$$= 51$$

$$(123)_{10} \rightarrow (?)_{10}$$

$$0 \times 10 + 1$$

$$= 1 \times 10 + 2$$

$$= 12 \times 10 + 3$$

$$= 123$$

$$12345$$

$$\text{base-10} : 0 \text{ to } 9 \rightarrow \text{decimal}$$

$$\Rightarrow \text{base-2} : 0 \text{ to } 1 \rightarrow \text{binary}$$

$$\text{base-8} : 0 \text{ to } 7 \rightarrow \text{octal}$$

$$\text{base-16} : 0 \text{ to } 9 \rightarrow \text{hexa-decimal}$$

$$(7AF)_{16}$$

$$(1100)_2 \rightarrow (12)_{10}$$

$$(111)_2 \rightarrow (7)_{10}$$

$$(10)_2 \rightarrow (2)_{10}$$

$$(1010)_2 \rightarrow (10)_{10}$$

$$(11)_2 \rightarrow (3)_{10}$$

$$(110011)_2 \rightarrow (51)_{10}$$

$$(1111)_2 \rightarrow (15)_{10}$$

$$112 \quad 64 \quad 32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1$$

$$5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 0$$

$$1 \quad 1 \quad 0 \quad 0 \quad 1 \quad 1$$

$$2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$$

$$32 + 16 + 0 + 0 + 2 + 1 = 51$$

$$128 \quad 64 \quad 32 \quad 16$$

$$1 \quad 1 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0$$

$$= 8421$$

$$128 + 64 + 8 + 4 = 204$$

$$(123)_{10} \xrightarrow[\text{??}]{\text{reverse}} (321)_{10}$$

$$\begin{aligned} & 0 \times 10 + 3 \\ = & 3 \times 10 + 2 \\ = & 32 \times 10 + 1 \\ = & 321 \end{aligned}$$

➤ Converting a number from **base-10** to **base-2**

$$(42)_{10} \longrightarrow (?)_2$$

2	42	0
2	21	1
2	10	0
2	5	1
2	2	0
2	1	1
	0	

$$(101010)_2$$

$$(27)_{10} \rightarrow ()_2 \quad 11011$$

$$(164)_{10} \rightarrow ()_2 \quad 10100100$$

$$(71)_{10} \rightarrow ()_2 \quad 1000111$$

$$(649)_{10} \rightarrow ()_2$$

2	649	1
2	324	0
2	162	0
2	81	1
2	40	0
2	20	0
2	10	0
2	5	1
2	2	0
2	1	1
	0	

$$(1010001001)_2$$

Power of 2's

2 ⁰	2 ¹	2 ²	2 ³	2 ⁴	2 ⁵	2 ⁶	2 ⁷	2 ⁸	2 ⁹	2 ¹⁰	...
1	2	4	8	16	32	64	128	256	512	1024	...

$$(42)_{10} \longrightarrow (101010)_2$$

^{5th} ^{4th} ^{3rd} ^{2nd} ^{1st} ^{0th}
₌ ₌ ₌ ₌ ₌ ₌

$$42 = 32 + 8 + 2$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ 2^5 & 2^3 & 2^1 \\ = & = & = \end{array}$$

$$(129)_{10} \rightarrow ()_2$$

$$129 = 128 + 1$$

$$\begin{array}{cc} \downarrow & \downarrow \\ 2^7 & 2^0 \end{array}$$

$$(547)_{10} \rightarrow ()_2$$

$$\frac{1}{76} \frac{0}{5} \frac{0}{4} \frac{0}{3} \frac{0}{2} \frac{0}{1} \frac{1}{0}$$

$$547 = 512 + 32 + 2 + 1$$

$$\begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ 2^9 & 2^5 & 2^1 & 2^0 \end{array}$$

$$\frac{1}{9} \frac{0}{8} \frac{0}{7} \frac{0}{6} \frac{1}{5} \frac{0}{4} \frac{0}{3} \frac{0}{2} \frac{1}{1} \frac{1}{0}$$

$$\begin{array}{l} (8421)_{10} \rightarrow 12 \\ (1100)_2 \rightarrow 12 \\ (01100)_2 \rightarrow 12 \\ (001100)_2 \rightarrow 12 \\ \vdots \end{array} \left. \begin{array}{l} \text{unsigned} \\ \text{integers} \end{array} \right\} (\geq 0)$$

Decimal Range of n-bit binary number

➤ How many n-bit binary numbers are possible ?

Handwritten notes and tables illustrating the calculation of the number of n-bit binary numbers.

Handwritten: 2^n

Handwritten: $\Rightarrow 2^n =$

Handwritten: $n=2$

Handwritten: $2^2 = 4$

(n-1) th	(n-2) th	...	2 nd	1 st	0 th
0/1			0/1	0/1	0/1

Handwritten: $n=3$

Handwritten: $8 = 2^3$

1 st	0 th
0	0
0	1
1	0
1	1

2 nd	1 st	0 th
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

➤ What is the **smallest** n-bit binary number ?

pos.	(n-1) th	(n-2) th	...	i th	...	2 nd	1 st	0 th
bit	0	0	...	0	...	0	0	0
wgt.	2 ⁿ⁻¹	2 ⁿ⁻²	...	2 ⁱ	...	2 ²	2 ¹	2 ⁰

Handwritten: $(0)_{10}$

➤ What is the **largest** n-bit binary number ?

pos.	(n-1) th	(n-2) th	...	i th	...	2 nd	1 st	0 th
bit	1	1	...	1	...	1	1	1
wgt.	2 ⁿ⁻¹	2 ⁿ⁻²	...	2 ⁱ	...	2 ²	2 ¹	2 ⁰

Handwritten: $()_{10}$

Handwritten: $2^0 + 2^1 + 2^2 + \dots + 2^{n-1}$

Handwritten: 2^0 is circled.

Handwritten: $2^0 = 1$

Handwritten: $\frac{2^n - 1}{2 - 1}$

Handwritten: $\frac{2^n - 1}{2 - 1} = (2^n - 1)_{10}$

Handwritten: $a = 2^0 = 1$

gp

$$\left. \begin{array}{l} a = 2^0 = 1 \\ r = 2 \\ \# = n \end{array} \right\} = (2^n - 1)_{10}$$

With **n-bits**, we can form 2^n binary numbers in the decimal range **[0, 2^n-1]**

$$2^n \left\{ \begin{array}{l} \overbrace{000 \dots 0}^n \rightarrow (0)_{10} \\ \vdots \\ \underbrace{111 \dots 1}_n \rightarrow (2^n - 1) \end{array} \right.$$

$$\begin{array}{l} 000 \rightarrow 0 \\ \vdots \\ 111 \rightarrow 7 \end{array}$$

➤ What is the **least** number of bits required to represent a **decimal number**?

$$\begin{array}{l} (12)_{10} \rightarrow (1100)_2 \quad \boxed{4} \text{ bits} \\ (18)_{10} \rightarrow (10010)_2 \quad \boxed{5} \text{ bits} \end{array}$$

$12 \rightarrow 01100 \leftarrow 5 \text{ bits}$
 $12 \rightarrow 001100 \leftarrow 6 \text{ bits}$
 $\vdots$

i/p : $(x)_{10} \rightarrow (\quad)_2$
min-bits ??

We know that, with **n-bits**, we can form **2^n binary numbers** in the decimal range **$[0, 2^n - 1]$**

$$\begin{array}{l} \overbrace{000 \dots 0}^n \rightarrow 0 \\ \vdots \\ \overbrace{111 \dots 1}^n \rightarrow 2^n - 1 \end{array} \quad \left. \vphantom{\begin{array}{l} 000 \dots 0 \\ \vdots \\ 111 \dots 1 \end{array}} \right\}$$

$$0 \leq x \leq 2^n - 1$$

$$x \leq 2^n - 1$$

$$2^n \geq x + 1$$

take $\log_2$ b/s

$$\log_2 2^n \geq \log_2 (x+1)$$

$$n \geq \log_2 (x+1)$$

$$x = 12$$

$$n \geq \log_2 (12+1)$$

$$n \geq \log_2 13$$

$$\log_2 8 = \log_2 2^3 = 3 \quad n \geq \underline{3 \dots}$$

$$\log_2 16 = \log_2 2^4 = 4$$

$$n \geq \log_2 (18+1)$$

$$\log_2 16 = 4$$

$$\geq \log_2 19$$

$$\cdot \log_2 16 = 4$$

$$\cdot \log_2 32 = \log_2 2^5 = 5$$

$$\lfloor \rfloor \leftarrow \text{floor}$$

$$\geq \log_2$$

$$\nearrow 4 \dots$$

$\downarrow$ (ci)

5

$\rightarrow \underline{\underline{\text{ceil fn}}}$

$$\Rightarrow \left\lceil \log_2 (x+1) \right\rceil$$

Complementary Number System

It is used to perform subtraction using addition.

$$\begin{array}{l}
 x - y \\
 \swarrow \text{complement of } y \\
 \text{or} \\
 x + (-y) \\
 \swarrow \text{-ve repr. of } y \\
 \text{base-2} \left\{ \begin{array}{l} 1^s \text{ complement}^* \\ 2^s \text{ complement} \checkmark \end{array} \right.
 \end{array}$$

$$\begin{array}{l}
 (12)_{10} \rightarrow (01100)_2 \\
 \begin{array}{c} \text{Seek bit} \\ \downarrow \\ (10011)_2 \leftarrow 1^s \text{ complement of } 12 \\ + 1 \\ \hline (10100)_2 \leftarrow 2^s \text{ complement or } -12 \text{ of } 12 \end{array}
 \end{array}$$

$$\begin{array}{l}
 (9)_{10} \rightarrow (01001)_2 \\
 \begin{array}{c} \text{Seek bit} \\ \downarrow \\ (10110)_2 \leftarrow 1^s \text{ comp. of } 9 \\ + 1 \\ \hline (10111)_2 \leftarrow 2^s \text{ comp. of } 9 \text{ or } -9 \end{array}
 \end{array}$$

If we add 0 to the front of the binary corr. to an unsigned int, then it doesn't affect that int

If we add 1 to the front of the binary corr. to an signed int (-ve) then it doesn't affect that int

$$\begin{array}{l}
 x = 12 \\
 y = 9 \\
 \begin{array}{r} 12 \\ - 9 \\ \hline ? \end{array} \quad \begin{array}{r} 12 \\ + (-9) \\ \hline ? \end{array} \rightarrow \begin{array}{l} 01100 \checkmark (12) \\ 01001 (9) \\ \downarrow 2^s \text{ comp} \\ 10111 (-9) \checkmark \end{array} \\
 \begin{array}{r} 12 \\ - 9 \\ \hline ? \end{array} \quad \begin{array}{r} 12 \\ + (-9) \\ \hline ? \end{array} \rightarrow \begin{array}{r} 01100 \\ 01001 \\ \hline 10101 \\ + 10111 \\ \hline 00011 \Rightarrow 3 \end{array}
 \end{array}$$

$$\begin{array}{r} 28 \\ - 12 \\ \hline ? \end{array}$$

$$\begin{array}{r} 28 \\ + (-12) \\ \hline ? \end{array}$$

1. find binary of 28
2. find binary of -12
3. add them

$$(28)_{10} \rightarrow (?)_2$$

$$\begin{array}{cccccc}
 0 & 1 & 1 & 1 & 0 & 0 \\
 \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
 2^4 & 2^3 & 2^2 & 2^1 & 2^0 & 2^{-1}
 \end{array}$$

$$28 = 16 + 8 + 4$$

$$011100 (= 28)$$

?

?

$$(28)_{10} \rightarrow (?)_2 \quad \begin{array}{r} 011100 \\ \underline{43210} \end{array} \quad (= 28)$$

$$(12)_{10} \rightarrow (?)_2 \quad \begin{array}{r} 0011.00 \\ \downarrow 2^5 \text{ amp} \\ 110100 \end{array} \quad (= -12)$$

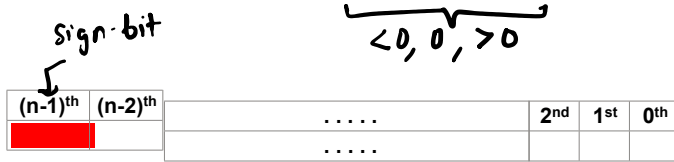
$$\begin{array}{r} \times 111 \\ 011100 \\ + 110100 \\ \hline 010000 \\ \hline \end{array} \Rightarrow 16$$

* left most of a signed bno. is sign-bit.

* if sign-bit is zero then it means the given signed bno. corr. to a ≥ 0 value

* if the sign-bit is one then it means the given signed bno. corr. to < 0 value

Binary Representation for Signed Integer



$$\Rightarrow (1 \ 0 \ 1 \ 0 \ 0)_2 \rightarrow (-ve)_{10}$$

$$\Rightarrow (0 \ 1 \ 1 \ 0)_2 \rightarrow (\geq 0)_{10}$$

➤ Converting a signed integer from base-2 to base-10

$$\Rightarrow (1 \ 0 \ 1 \ 0 \ 0)_2$$

$-2^4 \ 2^3 \ 2^2 \ 2^1 \ 2^0$

$$-16 + 0 + 4 + 0 + 0 = -12$$

$$\Rightarrow (0 \ 1 \ 1 \ 0)_2$$

$-2^3 \ 2^2 \ 2^1 \ 2^0$

$4 + 2 = 6$

$$\textcircled{1} \ 1 \ 0 \ 1 \ 0 \ 0$$

$-2^5 \ 2^4 \ 2^3 \ 2^2 \ 2^1 \ 2^0$

$$-32 + 16 + 4 = -12$$

$$\begin{array}{r} \hline 111 \dots 1 \ 10100 \\ \hline 27 \end{array}$$

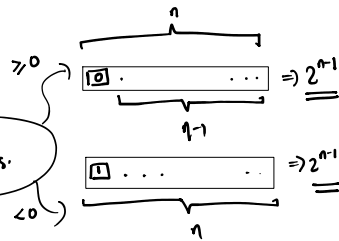
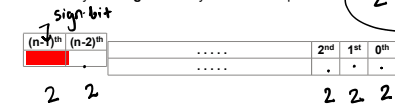
12

$$\begin{array}{r} \hline 0 \ 00 \dots 0 \ 1100 \\ \hline 28 \end{array}$$

$3 \ 2$

Decimal Range of n-bit Signed binary number

> How many n-bit signed binary numbers are possible?



> What is the **smallest** n-bit signed binary number given that **sign bit** is zero?

pos.	(n-1) th	(n-2) th	...	j th	...	2 nd	1 st	0 th
bit	0	0	...	0	...	0	0	0
wgt.	-2 ⁿ⁻¹	2 ⁿ⁻²	...	2 ^j	...	2 ²	2 ¹	2 ⁰

$\Downarrow$
(0)₁₀

> What is the **largest** n-bit signed binary number given that **sign bit** is zero?

pos.	(n-1) th	(n-2) th	...	j th	...	2 nd	1 st	0 th
bit	0	1	...	1	...	1	1	1
wgt.	-2 ⁿ⁻¹	2 ⁿ⁻²	...	2 ^j	...	2 ²	2 ¹	2 ⁰

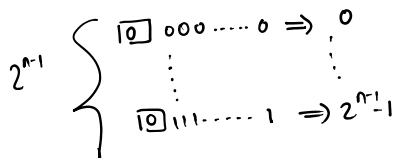
Handwritten calculation for the largest number:

$$2^0 + 2^1 + \dots + 2^{n-2}$$

AP: $a = 2^0 = 1$, $\lambda = 2$, $\# = n-1$

$$\frac{a(r^\# - 1)}{r - 1} = \frac{1(2^{n-1} - 1)}{2 - 1} = 2^{n-1} - 1$$

Assuming the sign bit is **zero**, with **n-bits**, we can form **2ⁿ⁻¹** signed binary numbers in the decimal range [0, 2ⁿ⁻¹-1].



> What is the **smallest** n-bit signed binary number given that **sign bit** is one?

pos.	(n-1) th	(n-2) th	...	j th	...	2 nd	1 st	0 th
bit	1	0	...	0	...	0	0	0
wgt.	-2 ⁿ⁻¹	2 ⁿ⁻²	...	2 ^j	...	2 ²	2 ¹	2 ⁰

$\Downarrow$
(-2ⁿ⁻¹)₁₀

> What is the **largest** n-bit signed binary number given that **sign bit** is one?

$\Rightarrow$

pos.	(n-1) th	(n-2) th	...	j th	...	2 nd	1 st	0 th
bit	1	1	...	1	...	1	1	1
wgt.	-2 ⁿ⁻¹	2 ⁿ⁻²	...	2 ^j	...	2 ²	2 ¹	2 ⁰

Handwritten calculation for the largest number with sign bit 1:

$$= -2^{n-1} + (2^{n-1} - 1)$$

$$= -2^{n-1} + 2^{n-1} - 1 = -1$$

in the binary of -1, all the bits are set

Assuming the sign bit is **one**, with **n-bits**, we can form **2ⁿ⁻¹** signed binary numbers in the decimal range [-2ⁿ⁻¹, -1].

$$2^{n-1} \left\{ \begin{array}{l} \boxed{1} 000 \dots 0 \Rightarrow -2^{n-1} \\ \vdots \\ \boxed{1} 111 \dots 1 \Rightarrow -1 \end{array} \right.$$

$$\eta \Rightarrow 2^n \text{ sbnos} \left\{ \begin{array}{l} \xrightarrow{\text{sb} \text{ bit} = 1} 2^{n-1} \text{ sbnos} \\ \xrightarrow{\text{sb} \text{ bit} = 0} 2^{n-1} \text{ sbnos} \end{array} \right.$$

$$\rightarrow \left\{ \begin{array}{l} \boxed{1} 000 \dots 0 \Rightarrow -2^{n-1} \Leftarrow -\infty \\ \vdots \\ \boxed{1} 111 \dots 1 \Rightarrow -1 \\ \boxed{0} 000 \dots 0 \Rightarrow 0 \\ \vdots \\ \boxed{0} 111 \dots 1 \Rightarrow 2^{n-1} \Leftarrow +\infty \end{array} \right.$$

$$\rightarrow [-2^{n-1}, 2^{n-1}]$$

**

-2^{31} to 2^{31}
INT_MIN

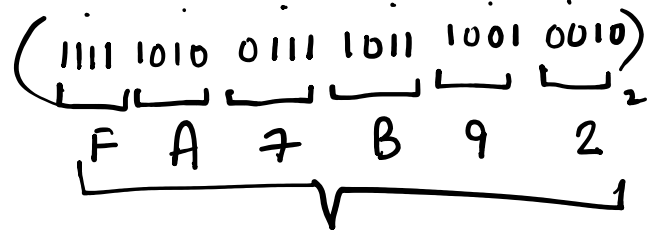
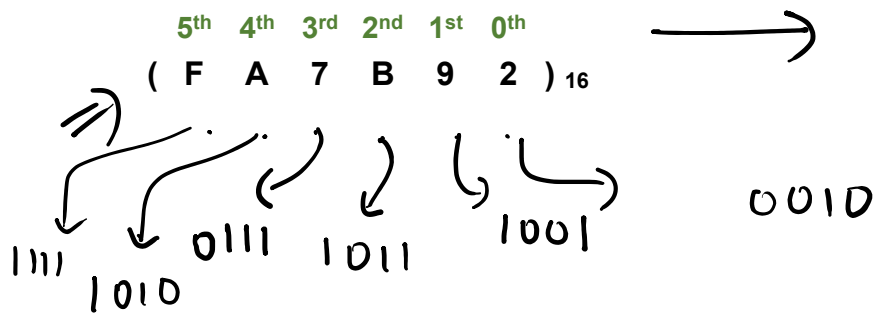
31
-1
INT-MAX

A-10	D-13
B-11	E-14
C-12	F-15
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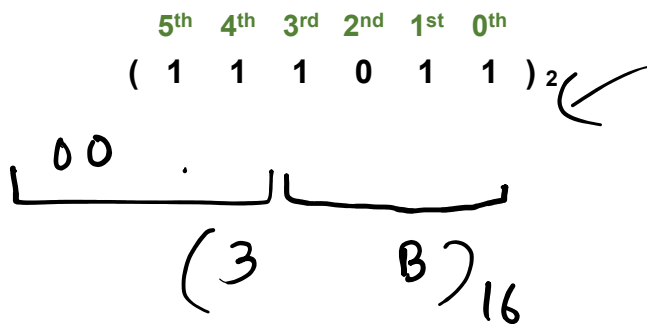
Hexa-Decimal (Base-16) Number System

In hexa-decimal number system, we form numbers using **16 digits** ranging from **0 to 9** & **A to F**.

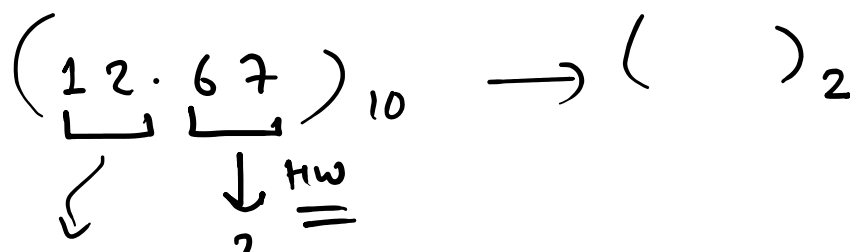
➤ Converting a number from **base-16** to **base-2**



➤ Converting a number from **base-2** to **base-16**



The hexa-decimal representation for a **n bit** binary number contains **n / 4** digits.



↙
1100

↓
?
.

hw
=

1100. _____